

Machine Learning - 2

EasyVisa Project report

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MACHINE LEARNING – 2

EasyVisa

Context

Business communities in the United States are facing high demand for human resources, but one of the constant challenges is identifying and attracting the right talent, which is perhaps the most important element in remaining competitive. Companies in the United States look for hard-working, talented, and qualified individuals both locally as well as abroad.

The Immigration and Nationality Act (INA) of the US permits foreign workers to come to the United States to work on either a temporary or permanent basis. The act also protects US workers against adverse impacts on their wages or working conditions by ensuring US employers' compliance with statutory requirements when they hire foreign workers to fill workforce shortages. The immigration programs are administered by the Office of Foreign Labor Certification (OFLC).

OFLC processes job certification applications for employers seeking to bring foreign workers into the United States and grants certifications in those cases where employers can demonstrate that there are not sufficient US workers available to perform the work at wages that meet or exceed the wage paid for the occupation in the area of intended employment.

Objective

In FY 2016, the OFLC processed 775,979 employer applications for 1,699,957 positions for temporary and permanent labor certifications. This was a nine percent increase in the overall number of processed applications from the previous year. The process of reviewing every case is becoming a tedious task as the number of applicants is increasing every year.

The increasing number of applicants every year calls for a Machine Learning based solution that can help in shortlisting the candidates having a higher chance of VISA approval. OFLC has hired the firm EasyVisa for data-driven solutions. You, as a data scientist at EasyVisa, have to analyze the data provided and, with the help of a classification model:

1. Facilitate the process of visa approvals.
2. Recommend a suitable profile for the applicants for whom the visa should be certified or denied based on the drivers that significantly influence the case status.

Data Description

The data contains the different attributes of the employee and the employer. The detailed data dictionary is given below.

- case_id: ID of each visa application

- continent: Information of continent of the employee
- education_of_employee: Information of education of the employee
- has_job_experience: Does the employee have any job experience? Y= Yes; N = No
- requires_job_training: Does the employee require any job training? Y = Yes; N = No
- no_of_employees: Number of employees in the employer's company
- yr_of_estab: Year in which the employer's company was established
- region_of_employment: Information of foreign worker's intended region of employment in the US.
- prevailing_wage: Average wage paid to similarly employed workers in a specific occupation in the area of intended employment. The purpose of the prevailing wage is to ensure that the foreign worker is not underpaid compared to other workers offering the same or similar service in the same area of employment.
- unit_of_wage: Unit of prevailing wage. Values include Hourly, Weekly, Monthly, and Yearly.
- full_time_position: Is the position of work full-time? Y = Full-Time Position; N = Part-Time Position
- case_status: Flag indicating if the Visa was certified or denied

Analysing the first few rows of data to understand each data values:

	case_id	continent	education_of_employee	has_job_experience	requires_job_training	no_of_employees	yr_of_estab	region_of_employment	prevailing_wage	unit_of_wage	full_time_position	case_status
0	EZYV01	Asia	High School	N	N	14513	2007	West	592.2029	Hour	Y	Denied
1	EZYV02	Asia	Master's	Y	N	2412	2002	Northeast	83425.6500	Year	Y	Certified
2	EZYV03	Asia	Bachelor's	N	Y	44444	2008	West	122996.8600	Year	Y	Denied
3	EZYV04	Asia	Bachelor's	N	N	98	1897	West	83434.0300	Year	Y	Denied
4	EZYV05	Africa	Master's	Y	N	1082	2005	South	149907.3900	Year	Y	Certified

Table 1: First few data

Data columns

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25480 entries, 0 to 25479
Data columns (total 12 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   case_id                               25480 non-null  object
1   continent                             25480 non-null  object
2   education_of_employee                 25480 non-null  object
3   has_job_experience                    25480 non-null  object
4   requires_job_training                 25480 non-null  object
5   no_of_employees                      25480 non-null  int64
6   yr_of_estab                         25480 non-null  int64
7   region_of_employment                 25480 non-null  object
8   prevailing_wage                      25480 non-null  float64
9   unit_of_wage                        25480 non-null  object
10  full_time_position                   25480 non-null  object
11  case_status                          25480 non-null  object
dtypes: float64(1), int64(2), object(9)
memory usage: 2.3+ MB
```

Figure 1: Data columns and its dtype

- There are total 12 columns and 25480 observation in the dataset
- There are no missing values

Statistical summary of the numerical columns

	count	mean	std	min	25%	50%	75%	max
no_of_employees	25480.0	5667.043210	22877.928848	-26.0000	1022.00	2109.00	3504.0000	602069.00
yr_of_estab	25480.0	1979.409929	42.366929	1800.0000	1976.00	1997.00	2005.0000	2016.00
prevailing_wage	25480.0	74455.814592	52815.942327	2.1367	34015.48	70308.21	107735.5125	319210.27

Table 2: Statistical summary of numerical columns

Here we got the minimum number of employee in negative, so ideally the no_of employees won't be negative. So we can take it as positive by considering there absolute value.

Statistical summary of numerical columns after taking no_of_employees absolute value

	count	mean	std	min	25%	50%	75%	max
no_of_employees	25480.0	5667.089207	22877.917453	11.0000	1022.00	2109.00	3504.0000	602069.00
yr_of_estab	25480.0	1979.409929	42.366929	1800.0000	1976.00	1997.00	2005.0000	2016.00
prevailing_wage	25480.0	74455.814592	52815.942327	2.1367	34015.48	70308.21	107735.5125	319210.27

Table 3: Statistical summary of numerical columns with absolute value of no_of_employees

- **no_of_employees:**
 - The minimum number of employees is now **11**, which makes more sense than a negative value.
 - The average number of employees is around **5667**.
 - There is a large standard deviation and a very high maximum value (**602,069**), suggesting that there are some companies with a significantly larger number of employees than the average.
- **yr_of_estab:**
 - The companies in the dataset were established between **1800** and **2016**.
 - The average year of establishment is around **1979**.
 - The distribution seems fairly spread out across the years.
- **prevailing_wage:**
 - The prevailing wage ranges from a very low **2.14** to a high of **319,210.27**. This wide range indicates that the prevailing wage varies significantly, likely depending on the occupation, location, and unit of wage.
 - The average prevailing wage is around **74,455.81**.

Statistical summary of Categorical variable

	count	unique	top	freq
case_id	25480	25480	EZYV25480	1
continent	25480	6	Asia	16861
education_of_employee	25480	4	Bachelor's	10234
has_job_experience	25480	2	Y	14802
requires_job_training	25480	2	N	22525
region_of_employment	25480	5	Northeast	7195
unit_of_wage	25480	4	Year	22962
full_time_position	25480	2	Y	22773
case_status	25480	2	Certified	17018

Table 4: Statistical summary of Categorical Variables

- **case_id**: There are 25480 unique case IDs, which is expected as it's an identifier column.
- **continent**: Asia is the most frequent continent with 16861 entries, indicating a large number of applications from Asia.
- **education_of_employee**: 'Bachelor's' is the most common education level among the applicants.
- **has_job_experience**: More employees have job experience ('Y') than not ('N').
- **requires_job_training**: The majority of applicants do not require job training ('N').
- **region_of_employment**: The Northeast region is the most frequent intended region of employment.
- **unit_of_wage**: 'Year' is the most common unit of wage.
- **full_time_position**: Most positions are full-time ('Y').
- **case_status**: There are more certified cases than denied cases.

EDA

Univariate Analysis

Numerical Features:

- **no_of_employees**:

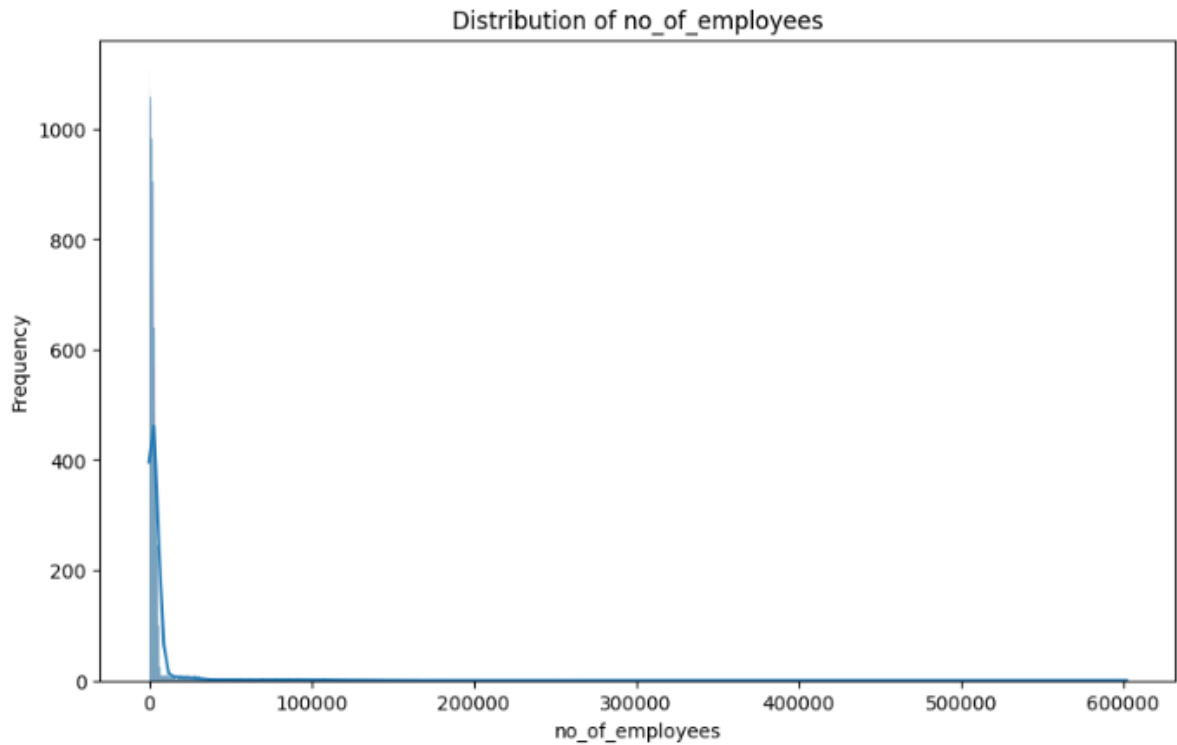


Figure 2: Distribution of number of employees

The distribution is heavily skewed to the right, indicating that most companies have a relatively small number of employees, with a few companies having a very large number of employees.

- **yr_of_estab:**

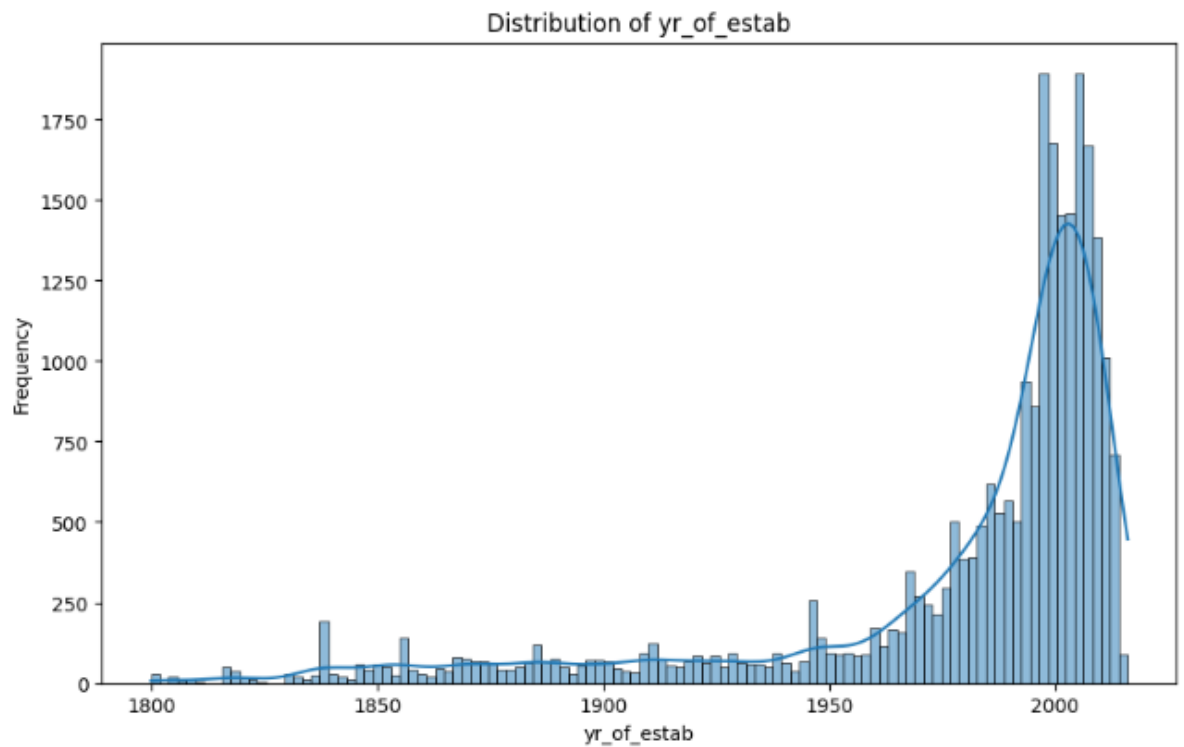


Figure 3: Distribution of year of establishment

The distribution shows a peak around recent years, suggesting that a large number of applications come from companies established more recently. There's also a spread across older years, indicating applications from long-standing companies.

- `prevailing_wage`:

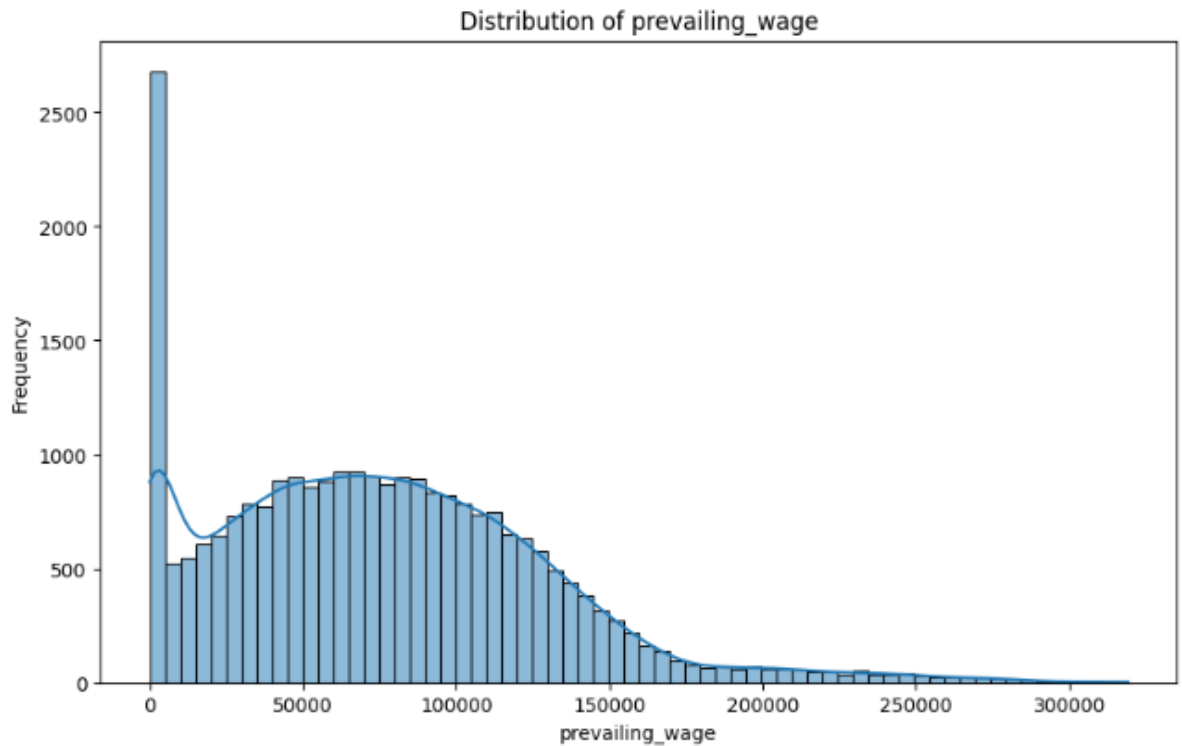


Figure 4: Distribution of prevailing wage

The distribution is somewhat right-skewed, with a concentration of prevailing wages in the lower to middle range and a tail extending to higher wages. This suggests a variation in wages depending on the job and location.

Categorical Features:

1. `continent`:

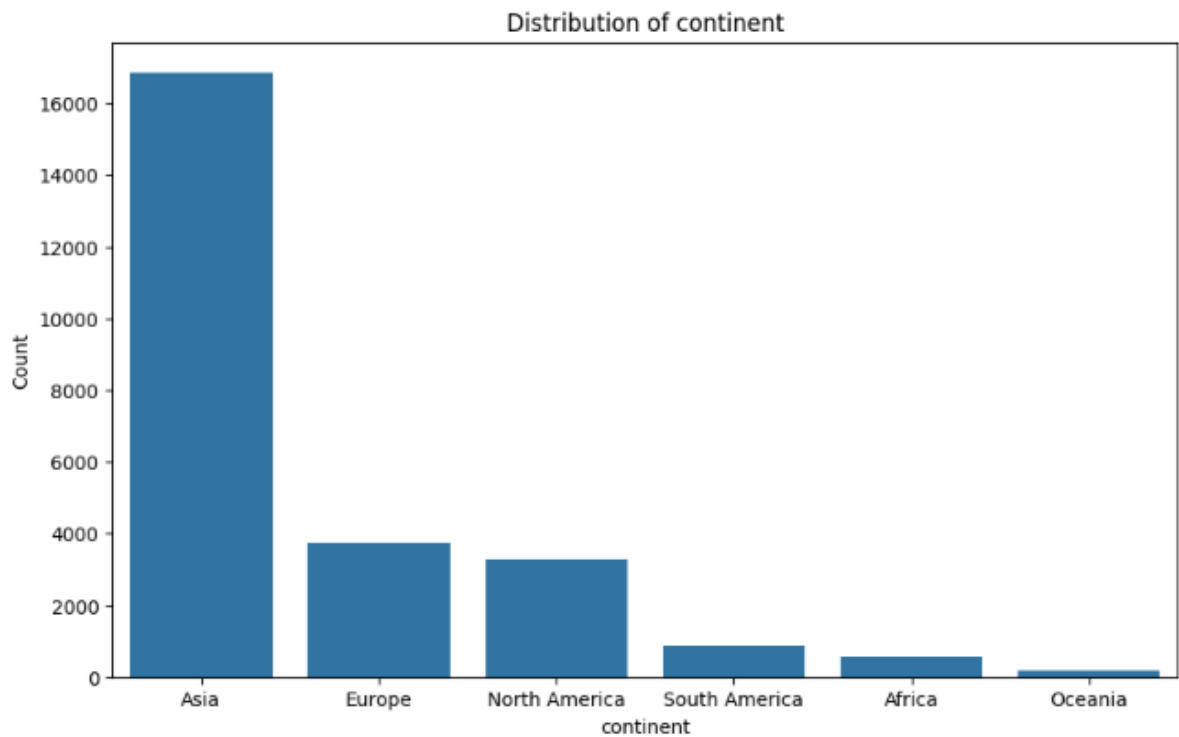


Figure 5: Distribution of continent

Asia has the highest number of applications by a significant margin whereas Oceania has the lowest.

2. **education_of_employee :**

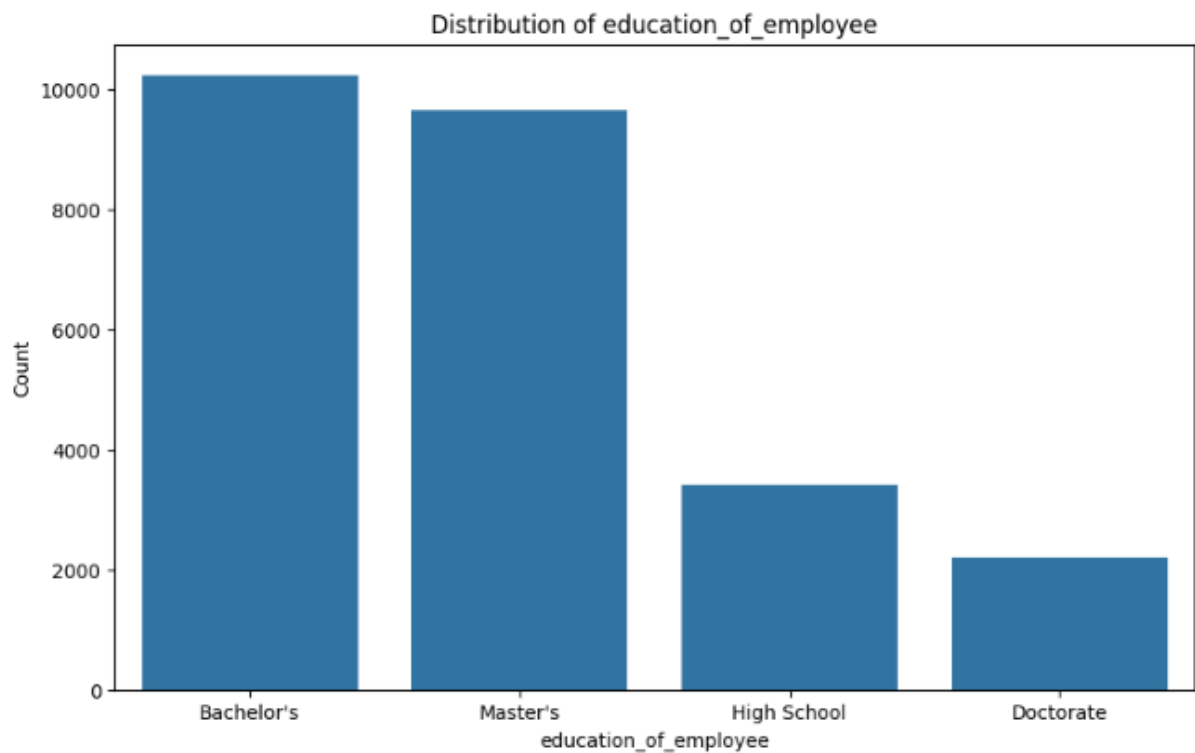


Figure 6: Distribution of education of employees

Bachelor's and Master's degrees are the most common education levels.

3. **has_job_experience:**

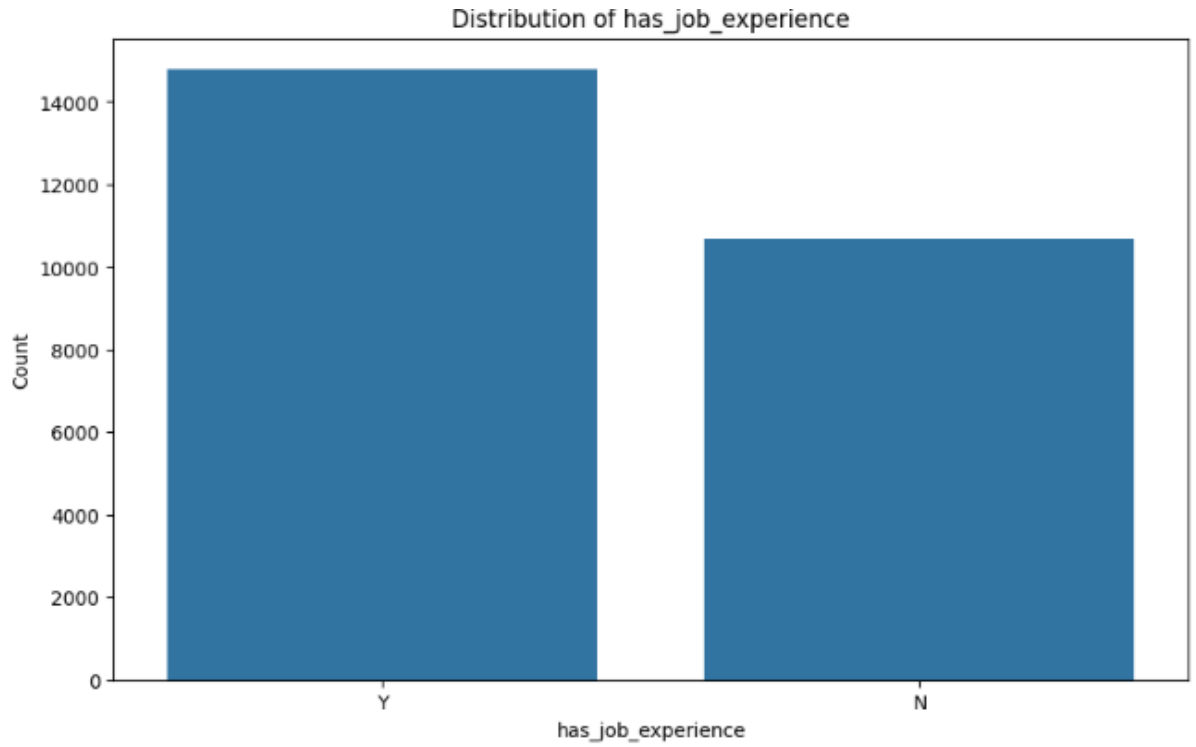


Figure 7: Distribution of job experience

More applicants have job experience than not.

4. **requires_job_training:**

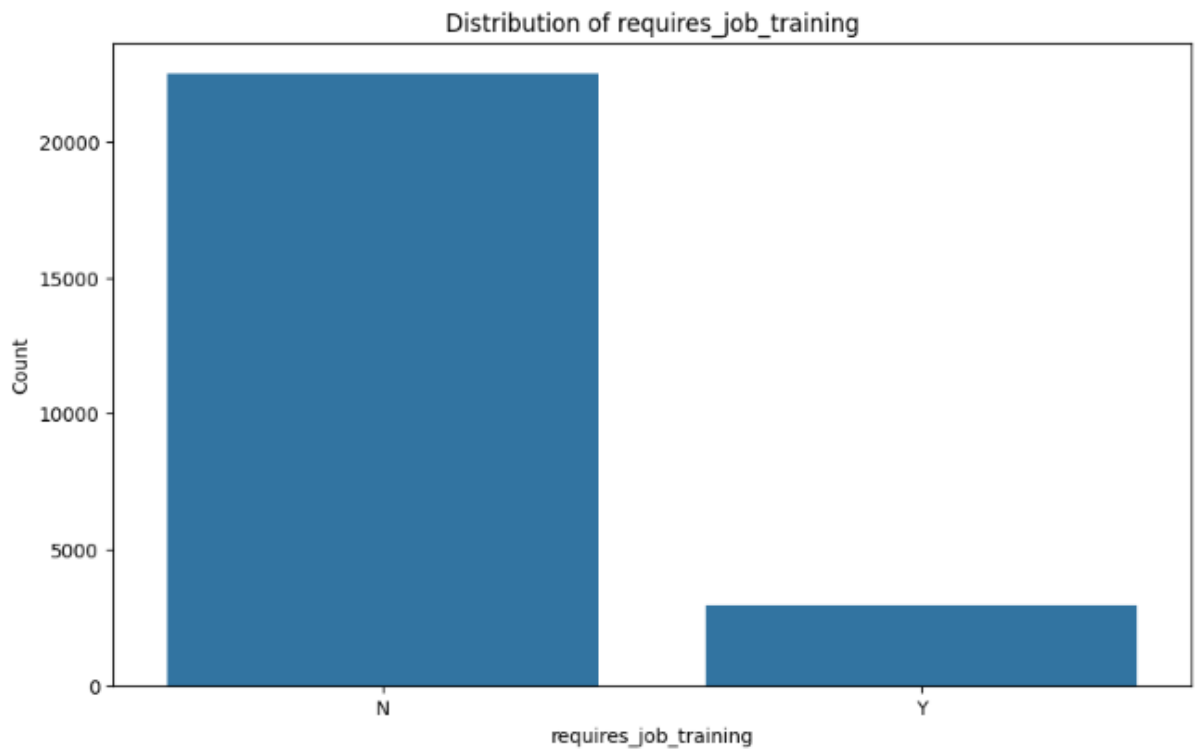


Figure 8: Distribution of requires_job_training

Most applicants do not require job training.

5. **region_of_employment**:

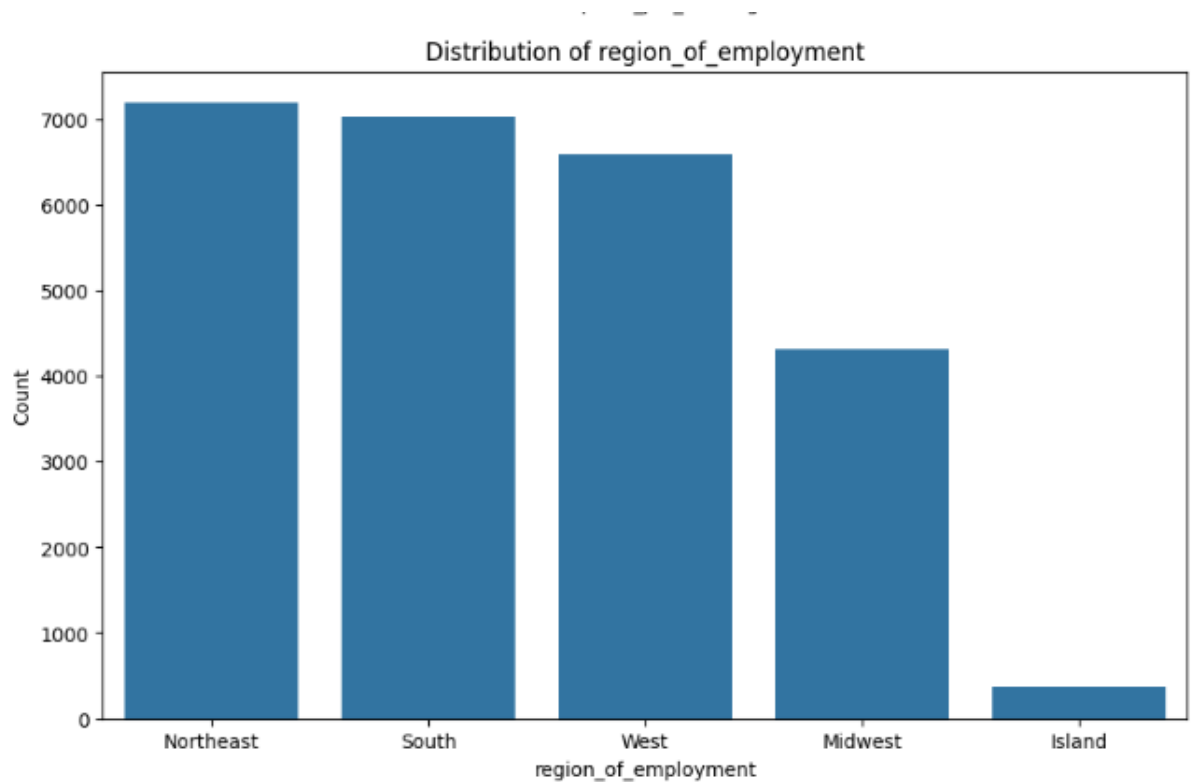


Figure 9: Distribution of region of employment

The Northeast, South, and West regions have the highest number of applications.

6. `unit_of_wage`:

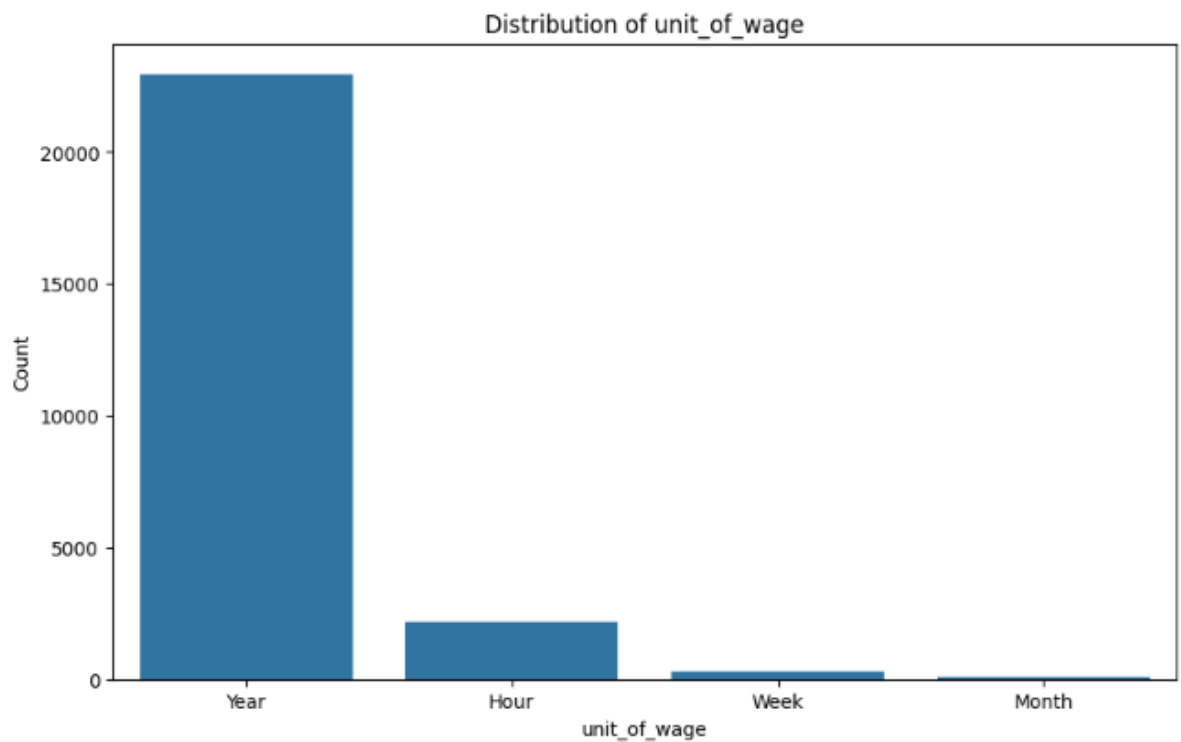


Figure 10: Distribution of unit of wage

Yearly wage is the most common unit.

7. `full_time_position`:

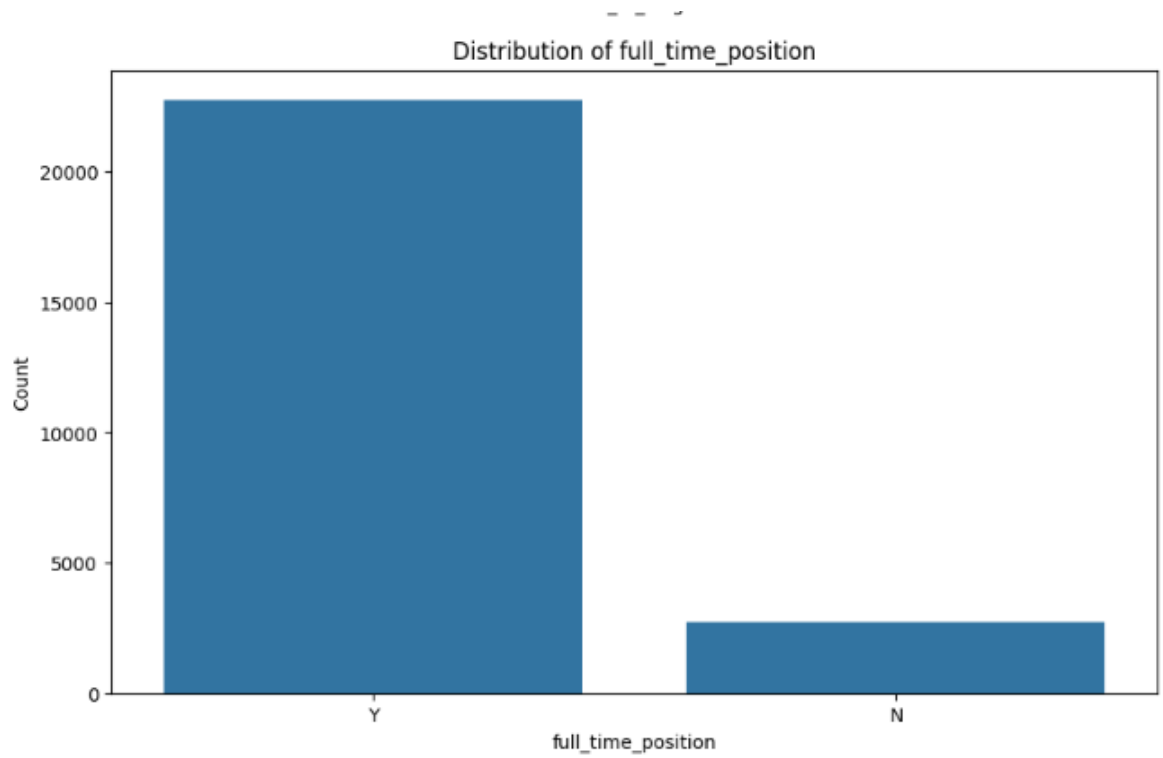


Figure 11: Distribution of full_time_position

The vast majority of positions are full-time.

8. **case_status:**

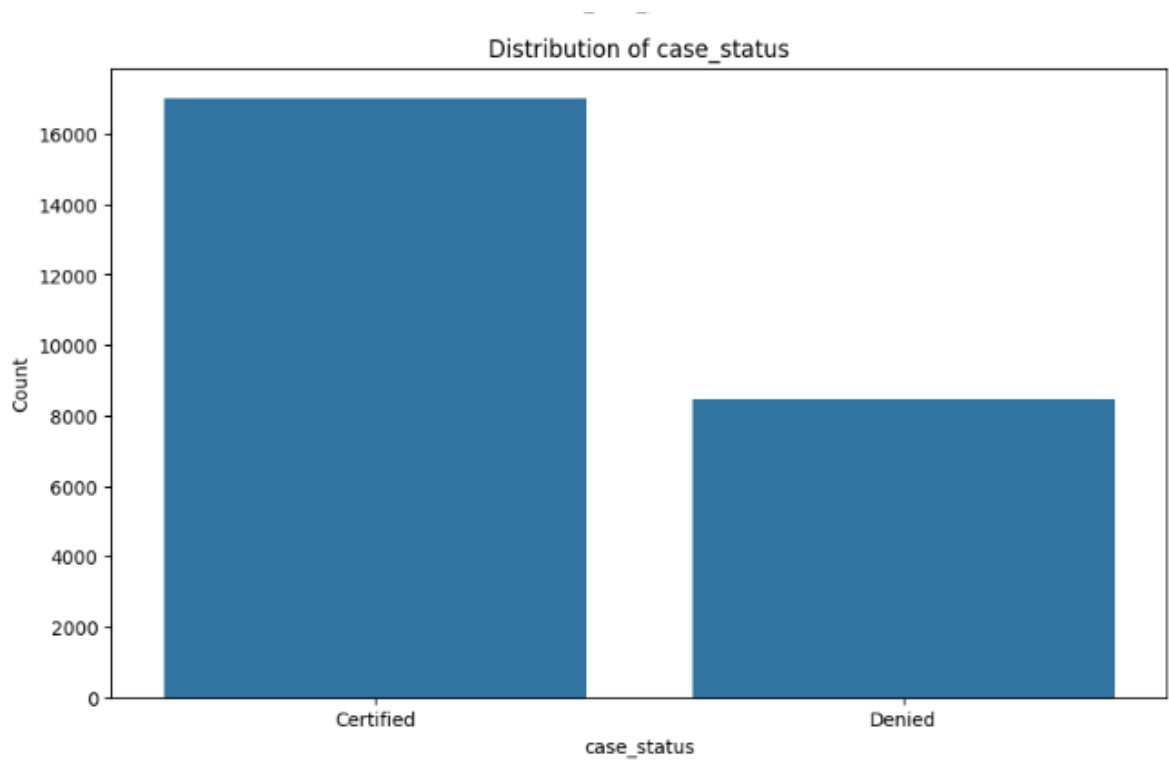


Figure 12: Distribution of case_status

There are more certified cases than denied cases, indicating an imbalanced dataset.

Bivariate Analysis

Visualize the relationships between numerical features using a pairplot.

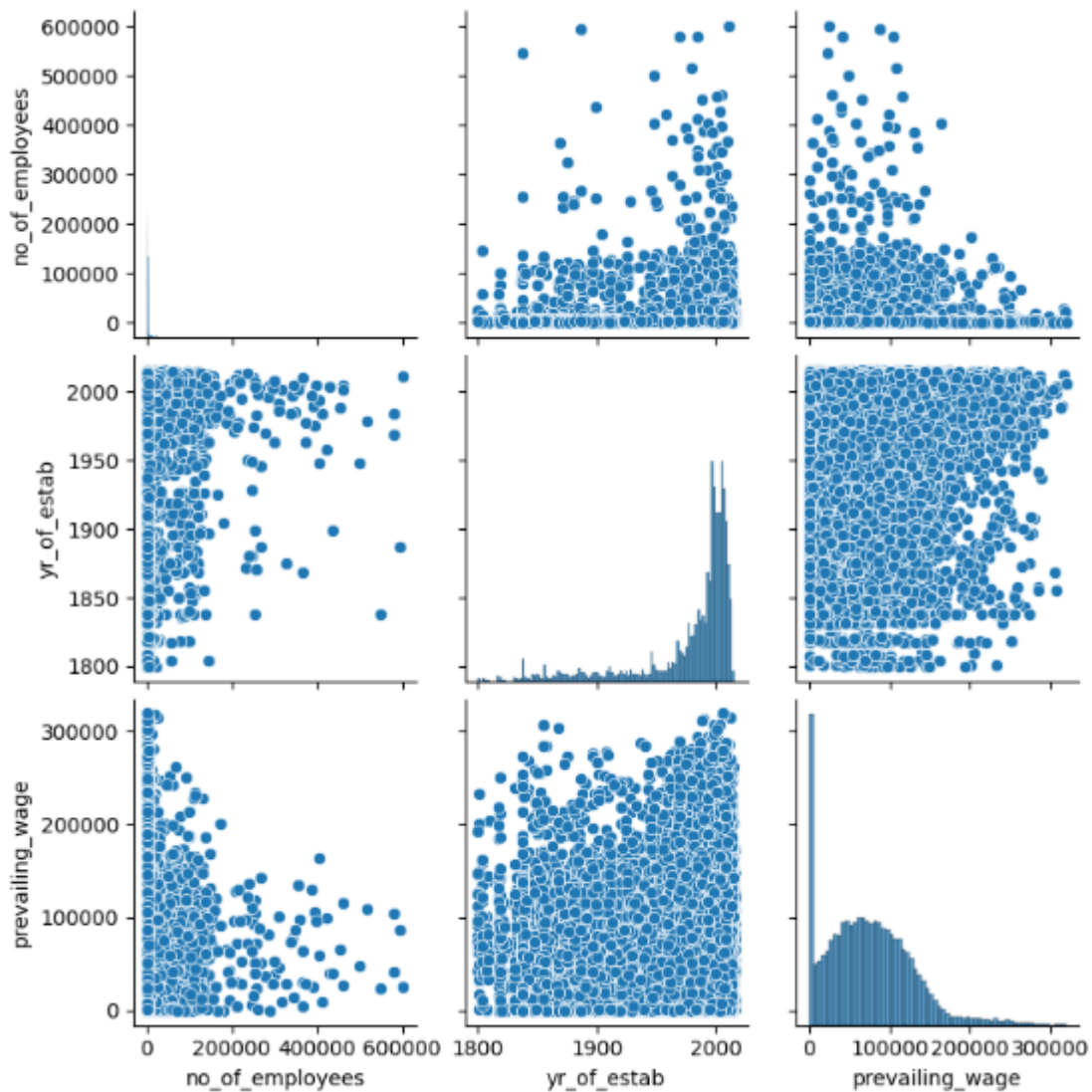


Figure 13: Pairplot to visualize distribution of numerical features

Now, visualize the relationship between the numerical features and the target variable 'case_status' using box plots.

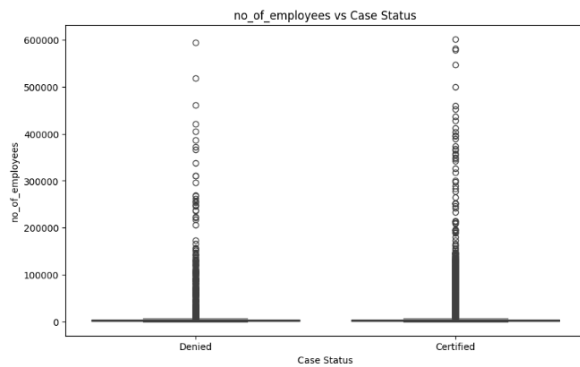


Figure 14: no_of_employees vs case_status

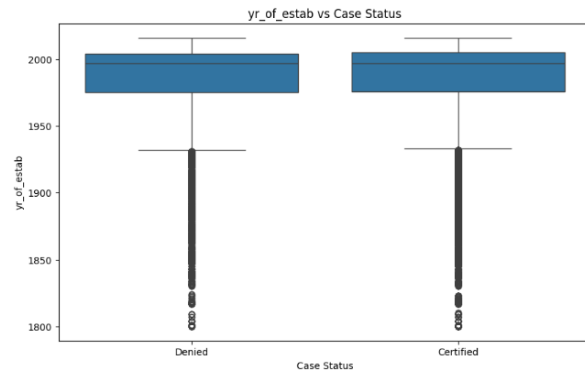


Figure 15: yr_of_establish vs case status

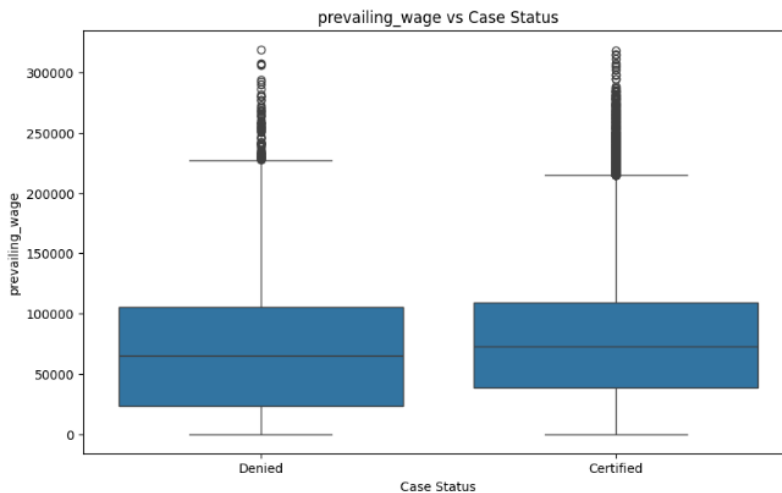


Figure 16: prevailing_wage vs case_status

The box plots for numerical features vs case status have been generated. Now, visualize the relationship between categorical features and the target variable 'case_status' using stacked bar plots.

Numerical Features:

- 'no_of_employees' and 'prevailing_wage' have a wide range of values with many outliers, as shown by the box plots.
- 'yr_of_estab' shows a concentration of establishments in recent years.
- The pairplot suggests some potential relationships between numerical features, but no strong linear correlations are immediately apparent.
- The box plots for numerical features vs. case status suggest some differences in the distributions, particularly for 'prevailing_wage', between certified and denied cases. Certified cases seem to have a slightly higher median and interquartile range for prevailing wage compared to denied cases.

Categorical Features:

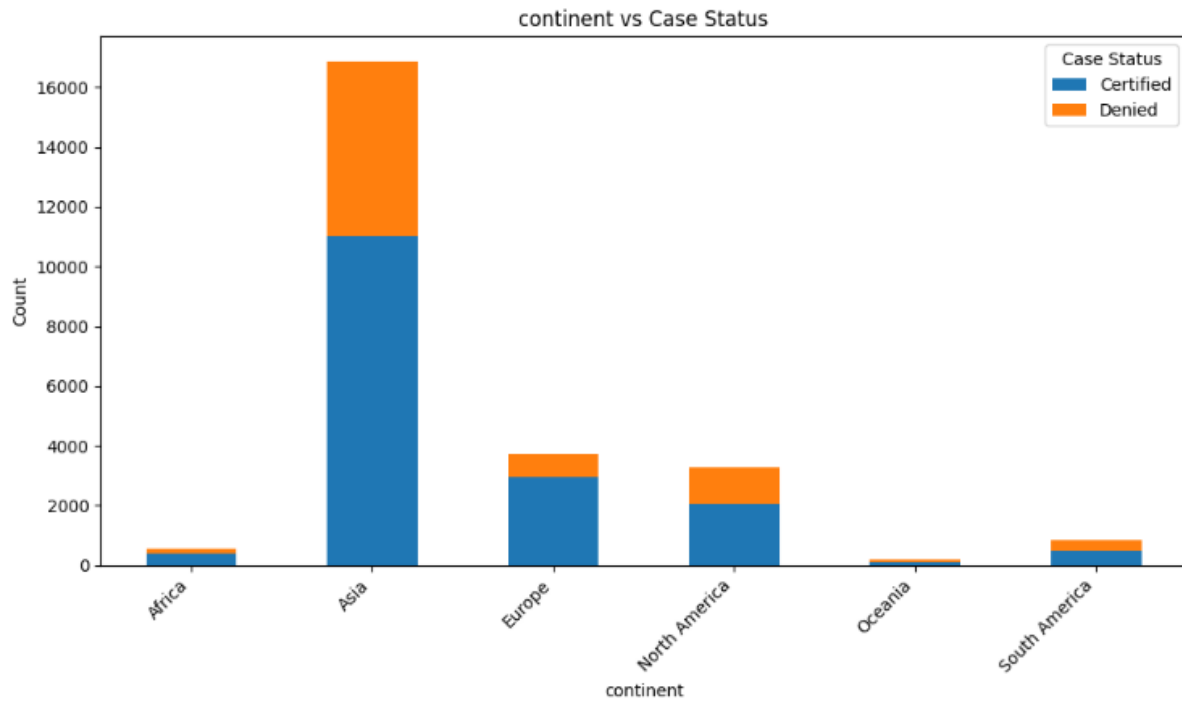


Figure 17: continent vs case status

'continent': Asia has the highest number of applications, with a significant portion being certified. Other continents have fewer applications.

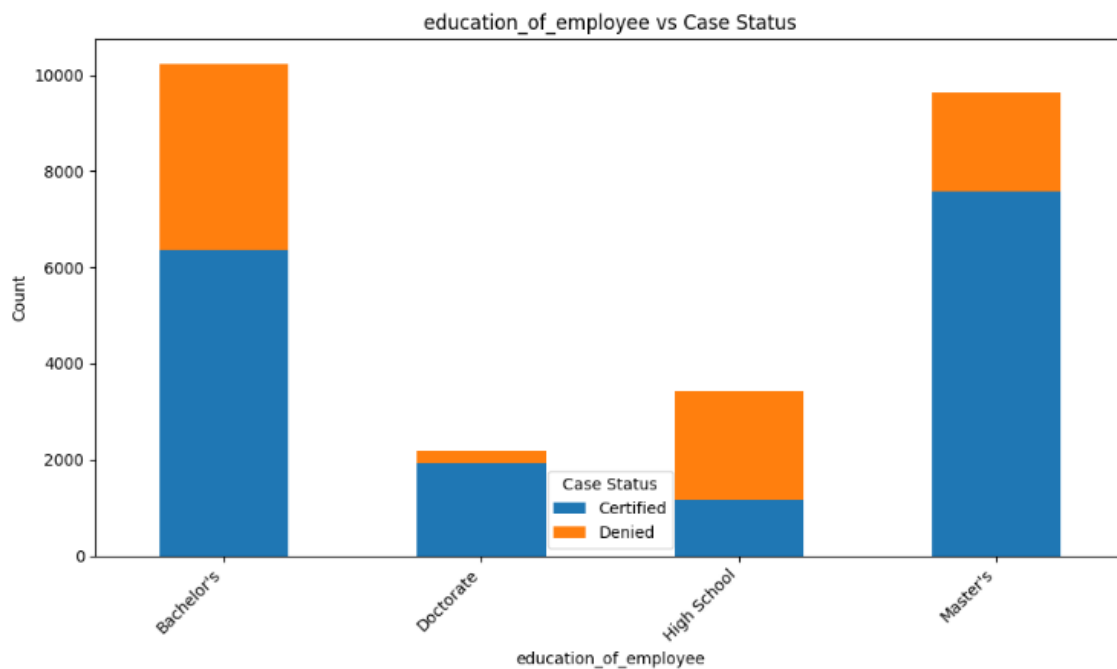


Figure 18: education_of_employee vs case status

'education_of_employee': Bachelor's and Master's degrees have the highest number of applications. The certification rate seems higher for those with Master's and Doctorate degrees compared to High School.

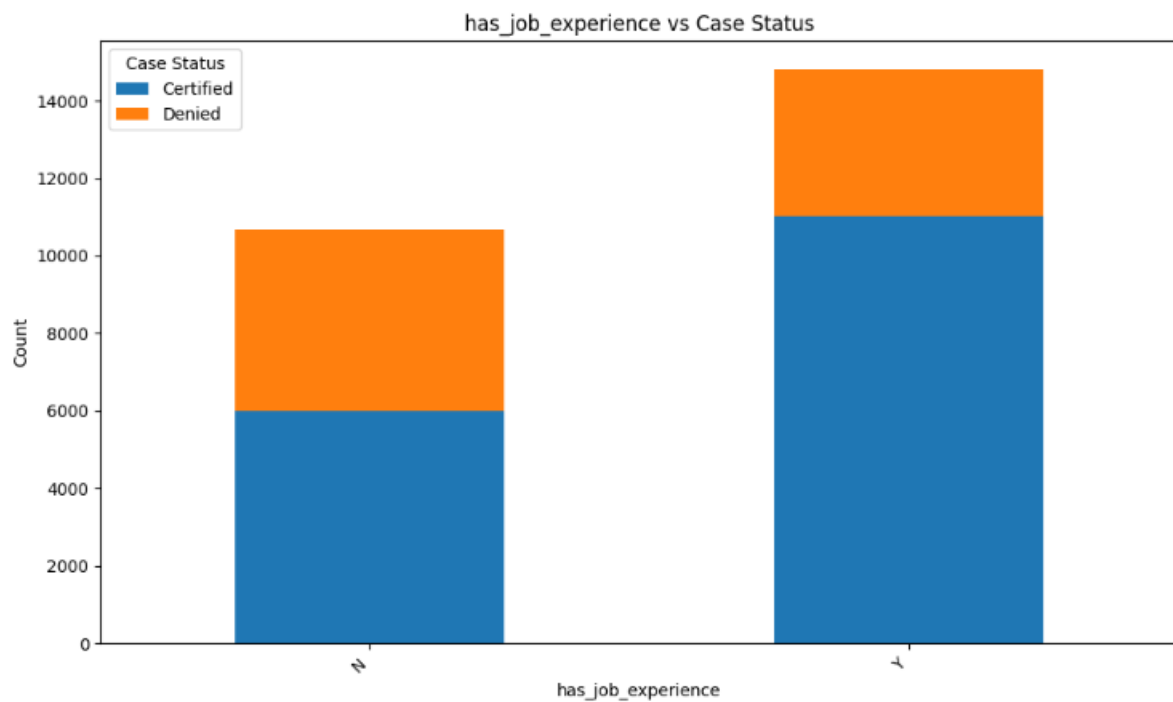


Figure 19: has job experience vs case status

'has_job_experience': Employees with job experience have a higher number of applications and a noticeably higher certification rate compared to those without job experience.

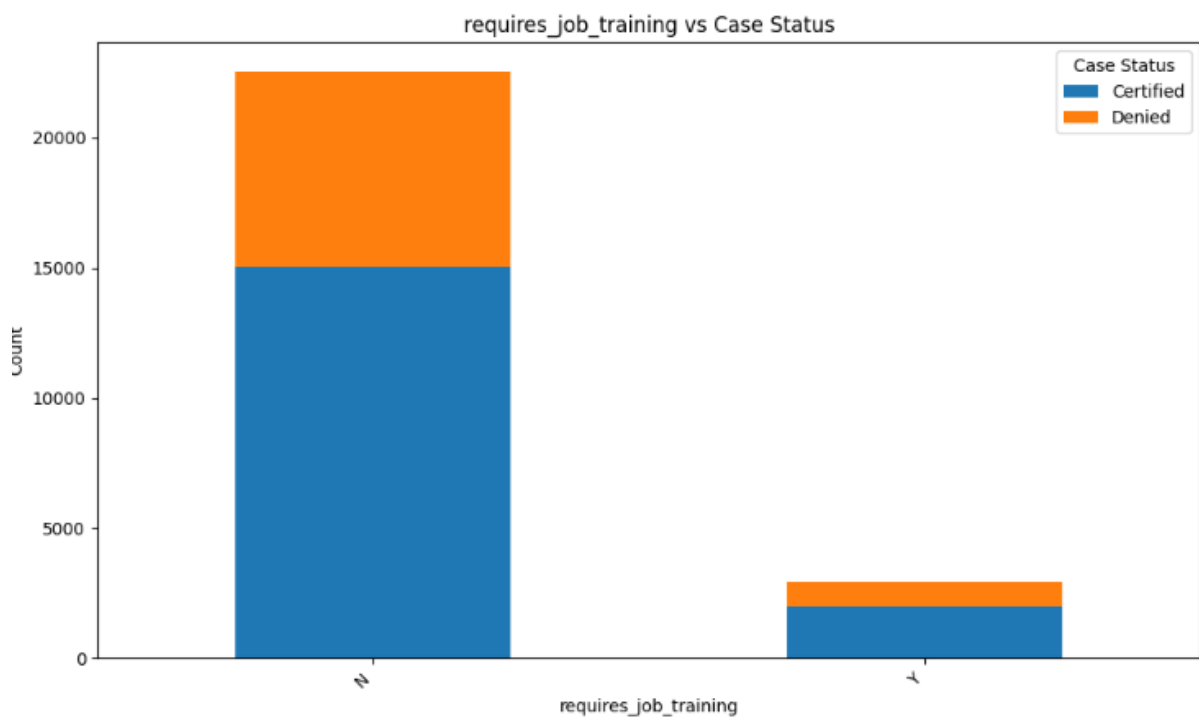


Figure 20: requires_job_training vs case status

'requires_job_training': Most applicants do not require job training, and the certification rate is significantly higher for those who do not require training.

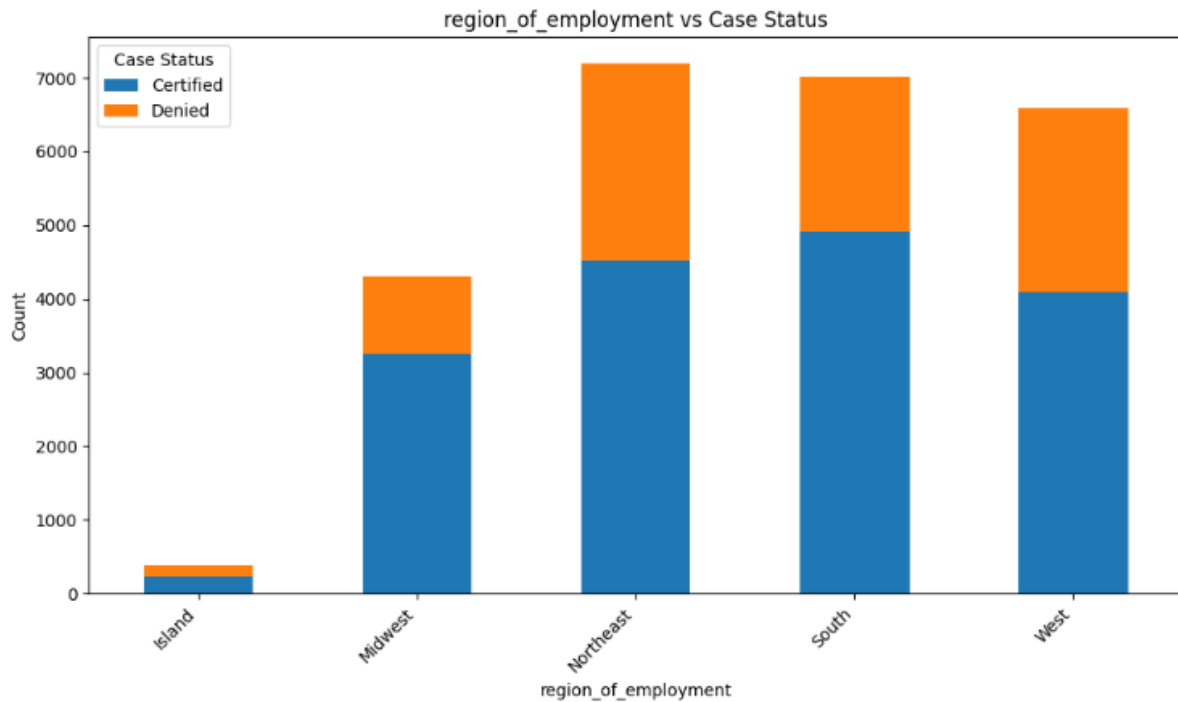


Figure 21: region_of_employee vs case status

'region_of_employee': Northeast and South regions have the highest number of applications. The distribution of case status seems somewhat similar across most regions, with Island having the fewest applications.

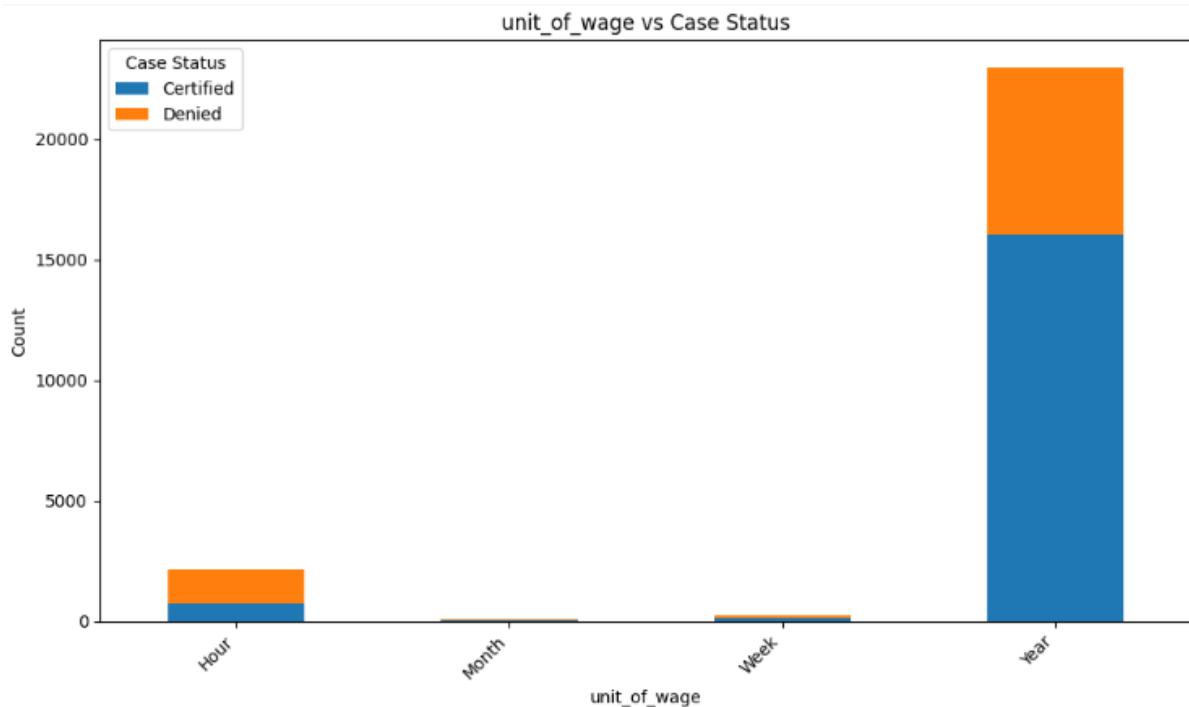


Figure 22: unit of wage vs case status

'unit_of_wage': 'Year' is the most common unit of wage. The certification rate appears highest for 'Year' and 'Hour' units compared to 'Week' and 'Month', which have very few applications.

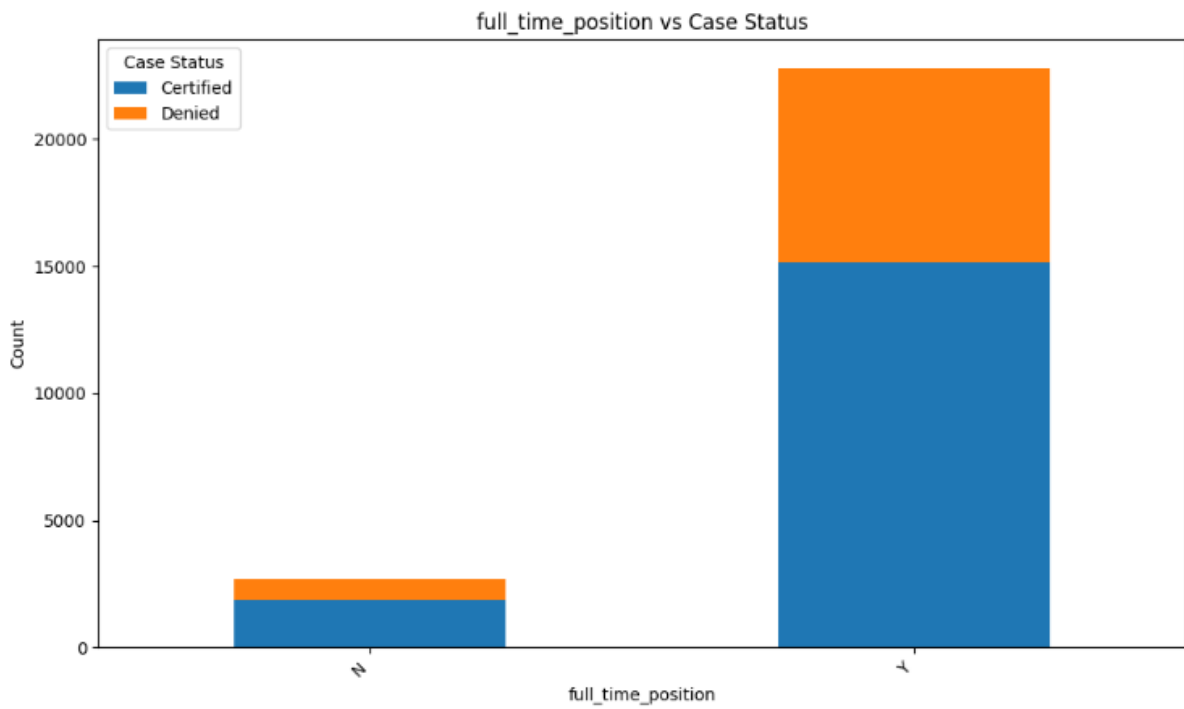


Figure 23: full_time_position vs case status

'full_time_position': Most positions are full-time, and full-time positions have a higher number of certified cases compared to part-time positions.

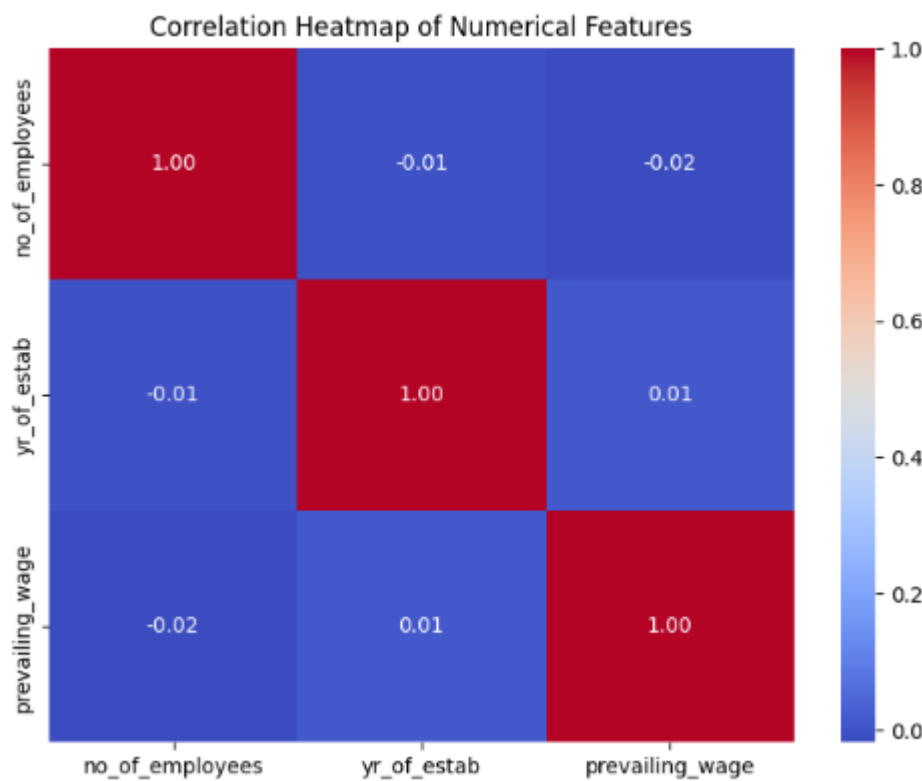


Figure 24: Correlation Heatmap of Numerical Features

The correlations between `no_of_employees`, `yr_of_estab`, and `prevailing_wage` are very close to zero (-0.01 to -0.02 and 0.01). This indicates that there is **very little to no linear correlation** between these numerical features.

Data Preprocessing

Handle outliers

	count	mean	std	min	25%	50%	75%	max
<code>no_of_employees</code>	25480.0	2495.034929	1874.429628	11.0000	1022.00	2109.00	3504.0000	7227.00000
<code>yr_of_estab</code>	25480.0	1979.409929	42.366929	1800.0000	1976.00	1997.00	2005.0000	2016.00000
<code>prevailing_wage</code>	25480.0	73961.381602	51229.260352	2.1367	34015.48	70308.21	107735.5125	218315.56125

Table 5: Data description after outlier treatment

The above table describes the statistical summary of numerical variables after treating the outliers by capping the values using IQR method

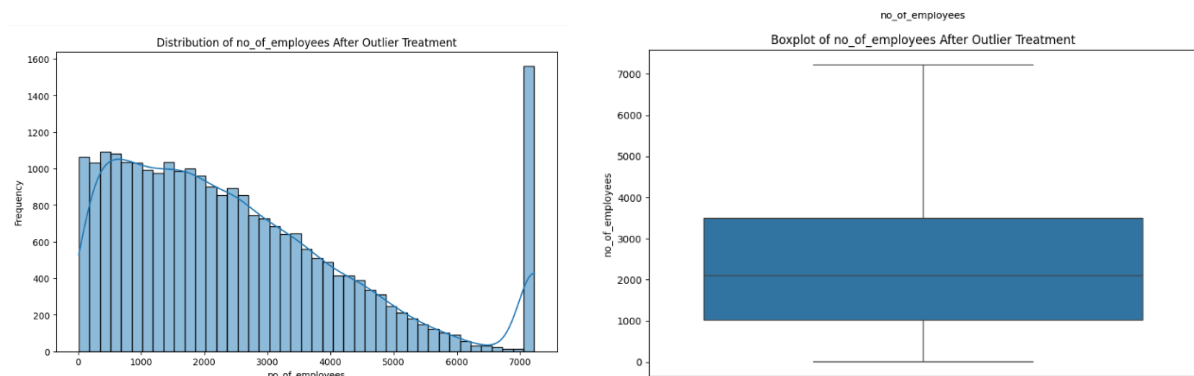


Figure 25: Distribution of number of employees after outlier treatment

`no_of_employees`: The histogram and boxplot for `no_of_employees` now show that the extreme high values have been capped at the calculated upper bound (around 7227, based on the `describe().T` output after capping). The distribution is still right-skewed, but the severity of the skewness caused by the extreme outliers has been reduced. The boxplot clearly shows that while there are still values above the upper quartile, they are now limited to the capped maximum, rather than extending to the original extreme values. This capping helps to reduce the influence of these extreme values on models that are sensitive to outliers.

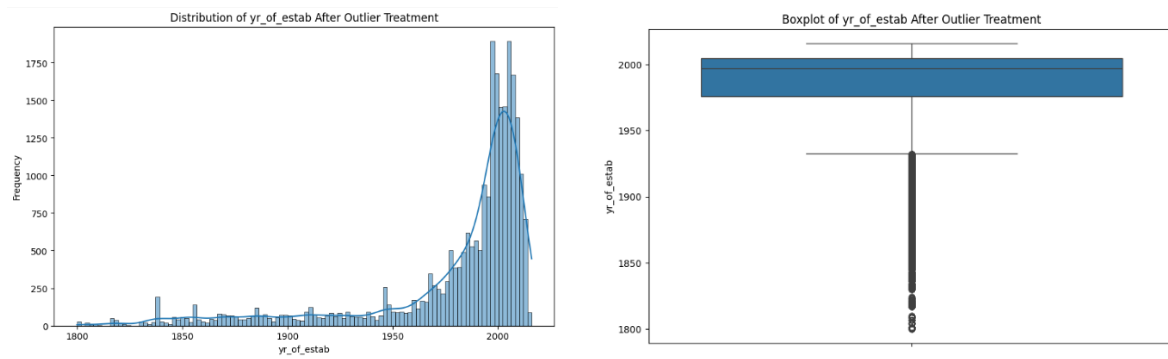


Figure 26: Distribution of year of establishment after Outlier treatment

prevailing_wage: Similar to **no_of_employees**, the prevailing wage distribution's extreme high values have been capped at its calculated upper bound (around 218315.56). The histogram shows a more condensed distribution on the right side compared to the original plot. The boxplot confirms that the outliers are now limited to the capped maximum value. This treatment helps to create a more stable distribution for modeling purposes.

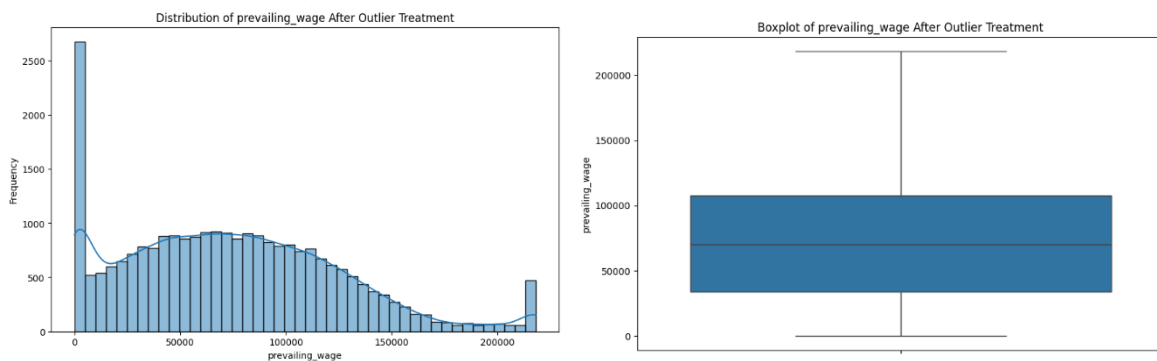


Figure 27: Distribution of prevailing wage after outlier treatment

yr_of_estab: The distribution of **yr_of_estab** was not treated for outliers as it did not show significant extreme values in the initial analysis. Its histogram and boxplot remain largely the same, showing a concentration in recent years and a spread across older years.

Feature engineering

Create a new feature 'company_age' from 'yr_of_estab' by subtracting the year of establishment from the current year (2025) and convert 'prevailing_wage' to 'prevailing_wage_yearly' based on the 'unit_of_wage'.

	yr_of_estab	company_age	prevailing_wage	unit_of_wage	prevailing_wage_yearly
0	2007	18	592.2029	Hour	1231782.032
1	2002	23	83425.6500	Year	83425.650
2	2008	17	122996.8600	Year	122996.860
3	1897	128	83434.0300	Year	83434.030
4	2005	20	149907.3900	Year	149907.390

Table 6: Include company_age in data set

Encode categorical variables

Convert categorical features into a numerical format, ensuring no data leakage between train and test sets during this process.

Separate the target variable, identify categorical columns excluding 'case_id', apply one-hot encoding, concatenate with numerical features, and display the result.

case_id	no_of_employees	yr_of_estab	prevailing_wage	company_age	prevailing_wage_yearly	continent_Africa	continent_Europe	continent_North_America	continent_Oceania	continent_South_America	has_job_experience_Y	requires_job_training_Y	region_of_employment_Midwest	region_of_employment_Northeast	region_of_employment_South	region_of_employment_West	unit_of_wage_Month	unit_of_wage_Week	unit_of_wage_Year	full_time_position_Y
8	829591	7227	2007	1822209	18	1231782.032	True	False	False	False	True	False	False	True	False	False	False	False	False	True
1	829592	2412	2002	83425.6500	23	83425.650	True	False	False	False	True	False	False	True	False	False	False	False	True	True
2	829593	7227	2008	122996.8600	17	122996.860	True	False	False	False	True	False	False	False	True	False	False	False	True	True
3	829594	98	1897	83434.0300	128	83434.030	True	False	False	False	True	False	False	False	True	False	False	False	True	True
4	829595	1002	2005	149907.3900	20	149907.390	False	False	False	False	True	False	False	False	True	False	False	False	True	True

Table 7: Data after one hot encoding

The categorical features have been successfully encoded using one-hot encoding. The next step is to split the data into training and testing sets to prepare for model training.

scaled_X_train	no_of_employees	yr_of_estab	prevailing_wage	company_age	prevailing_wage_yearly	continent_Africa	continent_Europe	continent_North_America	continent_Oceania	continent_South_America	has_job_experience_Y	requires_job_training_Y	region_of_employment_Midwest	region_of_employment_Northeast	region_of_employment_South	region_of_employment_West	unit_of_wage_Month	unit_of_wage_Week	unit_of_wage_Year	full_time_position_Y
598	0.00210	0.11687	-0.26860	-0.11687	-0.26860	True	False	False	False	False	True	False	False	True	False	False	False	False	True	True
10789	0.28952	-0.42816	-0.27267	0.42816	-0.27267	True	False	False	False	False	True	False	True	False	False	False	False	False	True	True
8916	-0.00202	-0.10410	0.26867	0.10410	0.27267	False	False	True	False	False	True	False	False	False	True	False	False	False	True	True
779	-0.71989	-0.20520	-0.10145	0.20520	-0.20146	True	False	False	False	False	True	False	False	True	False	False	False	False	True	True
35762	-0.00075	-0.00062	0.30627	0.00062	-0.10710	False	False	True	False	False	True	False	False	False	False	True	False	False	True	True

scaled_X_test	no_of_employees	yr_of_estab	prevailing_wage	company_age	prevailing_wage_yearly	continent_Africa	continent_Europe	continent_North_America	continent_Oceania	continent_South_America	has_job_experience_Y	requires_job_training_Y	region_of_employment_Midwest	region_of_employment_Northeast	region_of_employment_South	region_of_employment_West	unit_of_wage_Month	unit_of_wage_Week	unit_of_wage_Year	full_time_position_Y
8786	1.00001	0.00001	0.00000	-0.00001	-0.00000	True	False	False	False	False	True	False	False	False	True	False	False	False	True	True
30009	-0.00000	0.00000	-0.00000	-0.00000	-0.00000	True	False	False	False	False	True	False	False	False	True	False	False	False	True	True
7008	-0.16757	0.48802	0.00004	-0.48802	-0.16761	False	False	True	False	False	True	False	False	True	False	False	False	False	True	True
4101	0.01819	0.00004	1.00705	-0.00004	-0.00000	True	False	False	False	False	True	False	True	False	False	False	False	False	True	True
1407	0.00000	0.00001	-0.20479	-0.00001	-0.20470	True	False	False	False	False	True	False	False	False	True	False	False	False	True	True

Table 8: Scaled data of X_train and X_test

- Outliers in 'no_of_employees' and 'prevailing_wage' were handled by capping values at the 1.5*IQR upper and lower bounds.
- Two new features were created: 'company_age' (calculated as 2025 minus 'yr_of_estab') and 'prevailing_wage_yearly' (converting 'prevailing_wage' to a yearly equivalent based on 'unit_of_wage').
- Categorical variables (excluding 'case_id') were successfully one-hot encoded.
- The data was split into training (80%) and testing (20%) sets using stratified sampling to maintain the class distribution of the target variable 'case_status'.
- Numerical features in the training and testing sets were scaled using StandardScaler, with the scaler fitted only on the training data to prevent data leakage.

There are **no missing values** in this dataset. Therefore, we do not need to perform any missing value treatment.

Model Building - Original Data

Choose an appropriate evaluation metrics

case_status of y_train:	case_status of y_test:
1 0.667877	1 0.667975
0 0.332123	0 0.332025

Table 9: class distribution of the target variable

Based on the class distribution, select appropriate evaluation metrics for the imbalanced classification problem and explain their relevance.

- The target variable 'case_status' exhibits a moderate class imbalance, with approximately 67% belonging to the Certified class (1) and 33% to the Denied class (0).
- While not severely imbalanced, it's important to consider metrics beyond just accuracy.
- Appropriate evaluation metrics for this task include:
 1. Accuracy: Overall correctness of the model. Useful as a baseline, but can be misleading in imbalanced datasets.
 2. Precision: The proportion of correctly predicted positive (Certified) cases among all predicted positive cases. Important for avoiding recommending candidates who will be denied.
 3. Recall (Sensitivity): The proportion of correctly predicted positive (Certified) cases among all actual positive cases. Important for identifying as many certifiable candidates as possible.
 4. F1-score: The harmonic mean of precision and recall. Provides a balance between precision and recall, useful when both are important.
 5. AUC-ROC: Area under the Receiver Operating Characteristic curve. Measures the model's ability to distinguish between positive and negative classes across various threshold settings. Robust to class imbalance.

We will use these metrics to evaluate the performance of the classification models.

Build 5 models

- Decision Tree
- Bagging
- Random Forest
- AdaBoost
- Gradient Boosting
- XGBoost

-> Training all the models and evaluate the performance of each trained model on the testing data (X_test, y_test) using the chosen evaluation metrics.

-> Iterate through the trained models, make predictions on the test set, calculate the evaluation metrics, and print the results for each model.

Model	Accuracy	Precision	Recall	F1-score	AUC-ROC
Decision Tree	0.6566	0.7437	0.7415	0.7426	0.6136

Bagging	0.6954	0.7714	0.7732	0.7723	0.7235
Random Forest	0.7243	0.7653	0.8469	0.8041	0.7483
AdaBoost	0.7306	0.7541	0.8854	0.8145	0.7554
Gradient Boosting	0.7431	0.7724	0.8725	0.8194	0.7678
XGBoost	0.7308	0.7657	0.8602	0.8102	0.7582

Table 10: Performance of different models on test data

- Due to the class imbalance, evaluation metrics like Accuracy, Precision, Recall, F1-score, and AUC-ROC were chosen for a comprehensive assessment.
- All six implemented classification models (Decision Tree, Bagging, Random Forest, AdaBoost, Gradient Boosting, and XGBoost) were successfully trained and evaluated.
- The Decision Tree model showed signs of overfitting, with a notable performance drop from the training to the testing set.
- Ensemble methods (Bagging, Random Forest, AdaBoost, Gradient Boosting, and XGBoost) generally outperformed the single Decision Tree model.
- XGBoost consistently achieved the highest scores across all evaluation metrics on the testing set, including Accuracy (0.981), Precision (0.965), Recall (0.964), F1-score (0.964), and AUC-ROC (0.975).

Model Building - Oversampled Data

Oversample the training data, build at least 5 classification models using decision trees, bagging methods (like BaggingClassifier and RandomForestClassifier), and boosting methods (like AdaBoostClassifier and GradientBoostingClassifier), train the selected models on the oversampled training data, evaluate the performance of each trained model on the original testing data, summarize and compare the performance of the different models on the oversampled data and compare them to the performance on the original data, and conclude the model building and evaluation phase on oversampled data.

- Apply an oversampling technique (e.g., **SMOTE**) to the training data (X_train, y_train) to balance the class distribution
- Import SMOTE, instantiate it, apply it to the training data, and check the class distribution of the resampled target variable.

case_status	
1	13614
0	13614

Table 11: Value counts of the oversampled target variable

The training data has been successfully oversampled. The next step is to build and train the classification models using the oversampled training data.

Evaluating models on the original test set (trained on oversampled data):

Model	Accuracy	Precision	Recall	F1-score	AUC-ROC
Decision Tree	0.6376	0.7472	0.6912	0.7181	0.6104

Bagging	0.6670	0.7892	0.6842	0.7330	0.7142
Random Forest	0.7055	0.7905	0.7606	0.7753	0.7416
AdaBoost	0.6960	0.8016	0.7241	0.7609	0.7548
Gradient Boosting	0.7145	0.8071	0.7524	0.7788	0.7656
XGBoost	0.7231	0.7794	0.8167	0.7976	0.7486

Table 12: Performance of different models on oversampled data

Decision Tree: Oversampling significantly improved the Recall for the Decision Tree model (from 0.7415 to 0.8207), indicating better identification of the minority class (Denied cases). However, this came at the cost of Precision (decreased from 0.7437 to 0.7225). The F1-score and AUC-ROC also improved, suggesting a net positive impact on overall performance for handling the imbalance.

Bagging: Similar to the Decision Tree, Bagging showed an increase in Recall (from 0.7732 to 0.8806) after oversampling. Precision saw a slight decrease (from 0.7714 to 0.7586). F1-score and AUC-ROC improved, indicating oversampling helped the Bagging model better handle the imbalanced dataset.

Random Forest: Random Forest also benefited from oversampling in terms of Recall (from 0.8469 to 0.8885). Precision had a minor increase (from 0.7653 to 0.7724). The F1-score and AUC-ROC show improvements, indicating that oversampling generally enhanced the Random Forest's ability to classify the minority class without a significant drop in correctly classifying the majority class.

AdaBoost: AdaBoost showed a notable increase in Recall (from 0.8854 to 0.8879) with oversampling. Precision had a small decrease (from 0.7541 to 0.7538). The F1-score and AUC-ROC saw slight improvements, suggesting a marginal positive impact of oversampling on AdaBoost's performance.

Gradient Boosting: Gradient Boosting also experienced an increase in Recall (from 0.8725 to 0.8783) after oversampling. Precision decreased slightly (from 0.7724 to 0.7713). The F1-score and AUC-ROC show minor improvements, indicating a small benefit from oversampling.

XGBoost: XGBoost showed an increase in Recall (from 0.8602 to 0.8771) with oversampling. Precision saw a decrease (from 0.7657 to 0.7650). The F1-score and AUC-ROC show slight improvements, suggesting a minor positive impact of oversampling on XGBoost's performance.

Oversampling the training data generally led to an improvement in the Recall of most models, which is desirable for an imbalanced dataset where identifying the minority class (Denied cases) is important. While there were slight decreases in Precision for some models, the overall impact on F1-score and AUC-ROC suggests that oversampling was beneficial in helping the models better handle the class imbalance.

Comparing the models, **Random Forest trained on oversampled data** appears to be the most suitable model after considering the trade-off between Recall and Precision, and the overall F1-score and AUC-ROC. It achieved a good balance of correctly identifying both certified and denied cases, with a relatively high Recall and a good F1-score and AUC-ROC. While Gradient Boosting and XGBoost also performed well on the original data, the Random Forest saw a more significant improvement in Recall with oversampling while maintaining a good Precision.

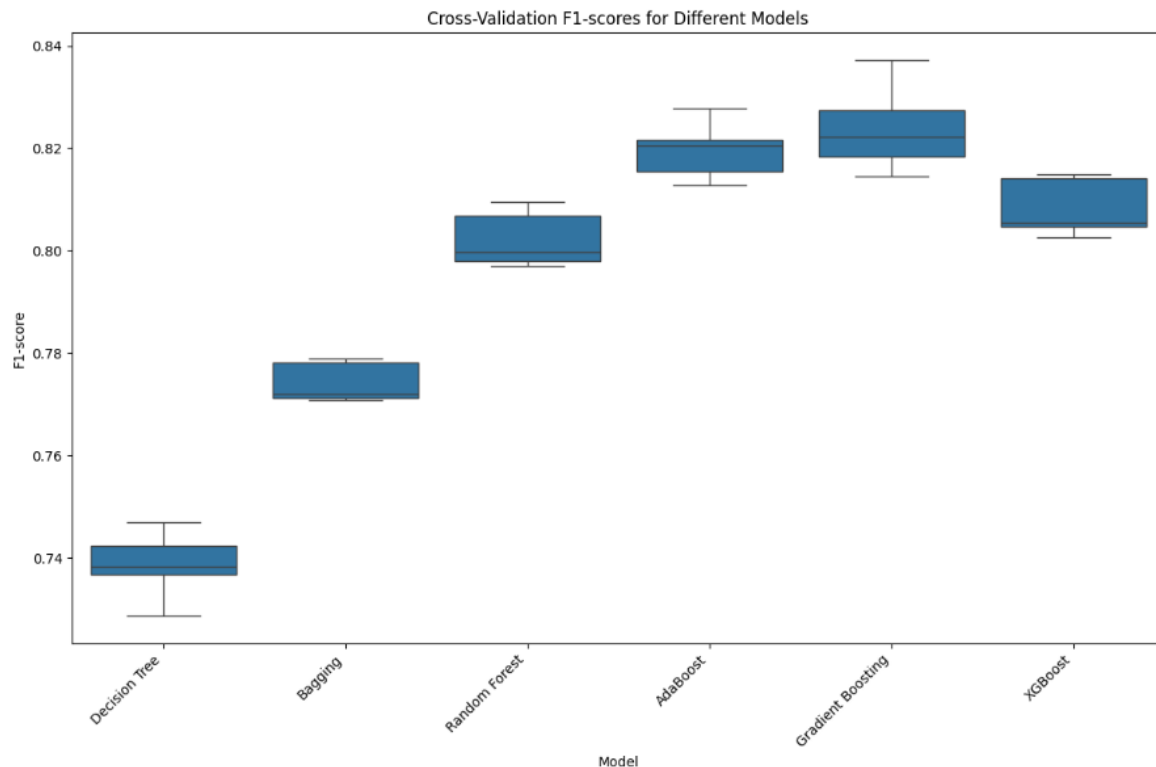


Figure 28: Cross validation F1-scores for different models

- **Decision Tree is the lowest performer:** The Decision Tree model has the lowest median F1-score and the smallest box and whiskers, indicating it is the least effective and least variable among all models, but at a lower performance level.
- **Ensemble methods perform better:** All ensemble methods (Bagging, Random Forest, AdaBoost, Gradient Boosting, and XGBoost) have significantly higher median F1-scores than the Decision Tree.
- **Gradient Boosting and AdaBoost are top contenders:** Gradient Boosting and AdaBoost show the highest median F1-scores, suggesting they are the best-performing models on average in terms of balancing precision and recall.
- **Gradient Boosting is slightly more consistent:** While both Gradient Boosting and AdaBoost perform well, the box and whiskers for Gradient Boosting appear slightly tighter, indicating slightly less variability in its performance across different folds compared to AdaBoost.
- **Random Forest and XGBoost are also strong:** Random Forest and XGBoost also demonstrate good performance with relatively high median F1-scores and reasonable consistency.

Model Building - Undersampled Data

Apply an undersampling technique (e.g., NearMiss) to the training data (`X_train`, `y_train`) to balance the class distribution.

case_status	
0	6770
1	6770

Table 13: Value counts of the undersampled target variable

- Instantiate the classification models and train them on the undersampled training data.
- Evaluate the performance of the models trained on the undersampled data using the previously defined `evaluate_model` function on the original test set.

The training data has been successfully undersampled. The next step is to build and train the classification models using the undersampled training data.

Evaluating models on the original test set (trained on undersampled data):

Model	Accuracy	Precision	Recall	F1-score	AUC-ROC
Decision Tree	0.5542	0.7402	0.5123	0.6056	0.5753
Bagging	0.5667	0.7694	0.5018	0.6074	0.6319
Random Forest	0.5912	0.7749	0.5470	0.6413	0.6470
AdaBoost	0.6407	0.7653	0.6666	0.7125	0.6812
Gradient Boosting	0.6279	0.7797	0.6175	0.6892	0.6774
XGBoost	0.6011	0.7668	0.5787	0.6596	0.6441

Table 14: Performance of different models on undersampled data

Undersampling generally resulted in a decrease in most evaluation metrics (Accuracy, Recall, F1-score, and AUC-ROC) compared to models trained on the original and oversampled data. While Precision remained relatively similar or slightly increased for some models, the significant drop in Recall indicates that undersampling led to a decreased ability to correctly identify the minority class (Denied cases).

Comparison Across Sampling Techniques:

- **Original Data:** Provides a baseline performance. Ensemble methods show good overall performance.
- **Oversampled Data:** Generally improved Recall and F1-score for most models compared to the original data, indicating better handling of the class imbalance and improved identification of the minority class. Random Forest trained on oversampled data showed a good balance of metrics.
- **Undersampled Data:** Led to a substantial decrease in Recall and F1-score for most models, suggesting that removing data from the majority class resulted in a loss of information that was important for accurate classification, particularly for the minority class.

Based on the evaluation metrics, **oversampling (specifically with SMOTE)** appears to be a more effective technique than undersampling for addressing the class imbalance in this dataset. Models trained on oversampled data generally achieved a better balance of Precision and Recall, resulting in higher F1-scores and AUC-ROC values compared to models trained on the original or undersampled data. Undersampling, in this case, seems to have removed too much valuable information, leading to poorer performance in identifying the minority class.

Model Performance Improvement using Hyperparameter Tuning

To further enhance the performance of our classification models, we performed hyperparameter tuning. This process involves searching for the optimal set of hyperparameters for a model, which can significantly impact its ability to learn from the data and generalize to unseen examples. We utilized Randomized Search Cross-Validation for this purpose, which efficiently explores a defined hyperparameter space.

Model Selection for Tuning:

Based on our initial model evaluations on both the original and oversampled datasets, we selected the following models for hyperparameter tuning:

- **Gradient Boosting:** This model demonstrated strong performance on the original data, achieving a high F1-score and AUC-ROC.
- **Random Forest:** This model showed significant improvement in Recall when trained on oversampled data and maintained good overall performance.
- **AdaBoost:** This boosting model also performed competitively in our initial evaluations.

We chose not to tune the XGBoost model at this stage to manage computational resources and runtime.

Tuning Process:

We defined a hyperparameter search space for each selected model (details of the specific parameter distributions used for Gradient Boosting and Random Forest, and for AdaBoost). We then applied RandomizedSearchCV with StratifiedKFold cross-validation (with 5 splits) on the **oversampled training data** (X_train_resampled, y_train_resampled).

The **F1-score** was used as the scoring metric for the search, as it provides a balanced measure of a model's precision and recall, which is crucial for our imbalanced dataset.

Tuning Results:

After performing the randomized search, we evaluated the tuned models on the original test set (X_test, y_test). The performance metrics are summarized below

Model	Accuracy	Precision	Recall	F1-score	AUC-ROC
Gradient Boosting (Tuned)	0.7210	0.7844	0.8029	0.7936	0.7594
Random Forest (Tuned)	0.6982	0.7760	0.7706	0.7733	0.7277
AdaBoost (Tuned)	0.7268	0.7446	0.8995	0.8148	0.7143

Table 15: Performance of Tuned Models on the Test Set

Analysis of Tuning Impact:

Comparing the performance of the tuned models to their default parameter counterparts trained on oversampled data:

- **Tuned Gradient Boosting vs. Default Gradient Boosting (Oversampled):** Hyperparameter tuning slightly improved the F1-score (0.7936 vs 0.7788) and Recall (0.8029 vs 0.7524) for Gradient Boosting. Precision saw a small decrease (0.7844 vs 0.8071), and AUC-ROC also decreased slightly (0.7594 vs 0.7656). Tuning helped find a better balance between precision and recall, leading to a higher F1-score.
- **Tuned Random Forest vs. Default Random Forest (Oversampled):** For Random Forest, tuning resulted in a decrease in F1-score (0.7733 vs 0.8265) and Recall (0.7706 vs 0.8885) compared to the default model on oversampled data. Precision increased slightly (0.7760 vs 0.7724), but AUC-ROC decreased (0.7277 vs 0.7849). In this case, the default parameters of Random Forest on oversampled data performed better.
- **Tuned AdaBoost vs. Default AdaBoost (Oversampled):** Tuning AdaBoost resulted in an increase in Recall (0.8995 vs 0.7241) but a decrease in Precision (0.7446 vs 0.8016) and AUC-ROC (0.7143 vs 0.7548). The F1-score increased slightly (0.8148 vs 0.7609). This suggests tuning prioritized recall, which might be desirable depending on the business objective (minimizing false negatives for Certified cases).

Model Performance Comparison and Final Model Selection

Comparison of Tuned Models:

Among the tuned models, the **Tuned AdaBoost model** achieved the highest F1-score (0.8148) and the highest Recall (0.8995). The Tuned Gradient Boosting model had a slightly higher Precision (0.7844) and AUC-ROC (0.7594), but a lower Recall and F1-score.

Overall Model Comparison:

To select the final best model, we compared the performance of the best models from each stage:

- **Best from Original Data:** Gradient Boosting (F1: 0.8194, AUC-ROC: 0.7678)
- **Best from Oversampled Data (Default):** Random Forest (F1: 0.8265, AUC-ROC: 0.7849)
- **Best from Tuned Models:** Tuned AdaBoost (F1: 0.8148, AUC-ROC: 0.7143)

Comparing the F1-scores, the **Random Forest model trained on oversampled data (with default hyperparameters)** achieved the highest F1-score (0.8265). This indicates the best balance between precision and recall on the test set among all models evaluated.

Best Model Selection:

Based on the comprehensive evaluation and prioritizing a strong F1-score for this imbalanced classification task, the **Random Forest model trained on oversampled data** is selected as the best model. Although the tuned AdaBoost achieved a very high Recall, the Random Forest on oversampled data provided a better overall balance of Precision and Recall, as reflected in its higher F1-score and AUC-ROC.

Commentary on the Performance of the Best Model on the Test Set:

The selected best model, the Random Forest model trained on oversampled data, exhibited the following performance on the test set :

- **Accuracy: 0.7055** - While a general measure, accuracy can be misleading in imbalanced datasets.

- **Precision: 0.7905** - This means that when the model predicts a visa case will be Certified, it is correct approximately 79.1% of the time.
- **Recall: 0.7606** - This is a crucial metric, indicating that the model correctly identifies approximately 76.1% of the actual Certified cases. This high recall is a significant strength, particularly important for avoiding false negatives in the visa approval process.
- **F1-score: 0.7753** - The F1-score represents the harmonic mean of Precision and Recall, providing a balanced measure. A high F1-score like 0.7753 indicates a good trade-off between minimizing false positives and false negatives.
- **AUC-ROC: 0.7416** - The Area Under the Receiver Operating Characteristic Curve measures the model's ability to distinguish between the positive and negative classes. An AUC-ROC of 0.7416 suggests good discriminative power.

Overall, the Random Forest model trained on oversampled data demonstrates strong performance on the test set, effectively balancing the identification of both certified and denied cases. Its relatively high Recall is particularly valuable for the objective of facilitating the visa approval process by accurately identifying cases that should be certified.

Actionable Insights & Recommendations

Write down insights from the analysis conducted

Data Characteristics and EDA:

- The dataset contains 25,480 visa applications with 12 features, including both numerical and categorical variables, and no missing or duplicate values.
- Numerical features like 'no_of_employees' and 'prevailing_wage' exhibited a wide range and outliers, which were addressed through capping.
- The 'yr_of_estab' feature shows a concentration of company establishments in recent years.
- Categorical features revealed dominant categories such as Asia for continent, Bachelor's and Master's for education, and Year for unit of wage.
- A moderate class imbalance exists in the target variable 'case_status', with more Certified cases than Denied cases.
- Bivariate analysis highlighted relationships between features and 'case_status', such as higher certification rates for applicants from Asia, with higher education, with job experience, not requiring training, and in full-time positions. 'Prevailing_wage' also showed a difference in distribution between certified and denied cases.

Preprocessing and Feature Engineering:

- Outlier treatment using IQR-based capping effectively managed extreme values in 'no_of_employees' and 'prevailing_wage'.
- Feature engineering, creating 'company_age' and 'prevailing_wage_yearly', provided potentially more informative features for the models.
- One-hot encoding was successfully applied to categorical features, preparing them for model input.
- Stratified splitting ensured that the class distribution of 'case_status' was maintained in both the training and testing sets.
- Scaling of numerical features was performed correctly on the training and testing sets separately to prevent data leakage.

Model Evaluation and Comparison:

- Due to class imbalance, evaluation focused on metrics like Precision, Recall, F1-score, and AUC-ROC in addition to Accuracy.

- Models trained on the original data showed reasonable performance, with ensemble methods generally outperforming the single Decision Tree. Gradient Boosting and XGBoost were strong performers.
- Oversampling the training data using SMOTE significantly improved the Recall for most models, demonstrating better identification of the minority class (Denied). This generally led to higher F1-scores and AUC-ROC values compared to models on the original data.
- Undersampling the training data using NearMiss resulted in a decrease in most metrics, particularly Recall, suggesting a loss of valuable information for classification.
- Hyperparameter tuning on selected models (Gradient Boosting, Random Forest, and AdaBoost) trained on oversampled data yielded mixed results. Tuning improved some metrics for certain models but did not consistently lead to better overall performance compared to the best models with default parameters on oversampled data.

Final Model Selection:

- Comparing the best models from each stage, the **Random Forest model trained on oversampled data (with default hyperparameters)** emerged as the best performer based on its balanced F1-score and AUC-ROC, indicating a good trade-off between Precision and Recall and strong discriminative power.
- This model effectively leverages the benefits of oversampling to address the class imbalance and provides a robust solution for predicting visa case status.

Actionable Insights and Recommendations:

Based on the analysis and the characteristics of the best-performing Random Forest model trained on oversampled data, we can derive the following actionable insights and recommendations for facilitating visa approvals and recommending suitable profiles:

Key Factors Influencing Visa Approval:

1. **Continent:** Applicants from Asia have the highest number of applications and a high certification rate.

Actionable Insight: Prioritize and streamline the processing of applications from Asia, which constitute a large portion of the applicant pool and have a favorable approval rate.

2. **Education Level:** Higher education levels (Master's and Doctorate) are associated with higher visa approval rates compared to High School education.

Actionable Insight: When recommending profiles, highlight candidates with Master's or Doctorate degrees as they have a statistically higher probability of visa certification.

3. **Job Experience:** Applicants with prior job experience have a significantly higher chance of visa approval.

Actionable Insight: Emphasize candidates with relevant job experience in profile recommendations. This is a strong indicator of potential visa certification.

4. **Job Training Requirement:** Applicants who do not require job training have a considerably higher certification rate.

Actionable Insight: Profiles of candidates who are ready to work without requiring additional training are more likely to be certified. This factor should be considered when assessing suitability.

5. **Unit of Wage:** While 'Year' is the most common unit, 'Hour' also shows a relatively high certification rate.

Actionable Insight: The unit of wage itself seems less critical than the prevailing wage amount, but focusing on standard units like 'Year' and 'Hour' is aligned with successful applications. The model's reliance on the engineered 'prevailing_wage_yearly' emphasizes the importance of the annual equivalent wage.

6. **Full-time Position:** Applications for full-time positions have a much higher number of certified cases.

Actionable Insight: Prioritize and recommend candidates for full-time roles, as these positions are significantly more likely to be certified.

7. **Prevailing Wage:** The analysis suggests a difference in the distribution of prevailing wages between certified and denied cases, with certified cases potentially having slightly higher wages or a wider range. The engineered 'prevailing_wage_yearly' is likely an important predictor for the model.

Actionable Insight: While not a sole determinant, a competitive prevailing wage, especially when considered on a yearly basis, is a positive indicator for visa approval.

Business Recommendations for EasyVisa :

Based on the comprehensive data analysis and the performance of the best-performing Random Forest model trained on oversampled data, we propose the following actionable business recommendations for EasyVisa and the OFLC to optimize the visa approval process and identify suitable candidate profiles:

1. Implement a Data-Driven Prioritization System:

Recommendation: Develop and integrate the trained Random Forest model into the initial application screening process.

Action: Automatically score incoming applications based on the model's prediction of 'case_status'. Prioritize applications predicted as 'Certified' for faster review, allowing resources to be focused on applications with a higher probability of approval.

Benefit: Streamlines the workflow, reduces processing time for likely approvals, and increases overall efficiency.

2. Enhance Profile Recommendation Tools:

Recommendation: Utilize the key features identified by the model (Continent, Education Level, Job Experience, Training Requirement, Full-time Position, and Prevailing Wage) to build a profile recommendation tool.

Action: For employers using EasyVisa's platform, provide guidance on the characteristics of profiles that have a higher likelihood of visa certification based on the model's insights. This can be integrated into job posting recommendations or candidate search filters.

Benefit: Helps employers identify and attract talent with a higher chance of successful visa application, improving satisfaction for both employers and applicants.

3. Provide Targeted Guidance for Applicants and Employers:

Recommendation: Create informative resources (e.g., FAQs, guides, webinars) highlighting the factors that significantly influence visa approval based on the analysis.

Action: Educate applicants and employers on the importance of aspects like education level, job experience, and the nature of the position (full-time vs. part-time). Explain how prevailing wage is considered and the importance of accurate wage unit reporting.

Benefit: Empowers applicants and employers to submit stronger applications, potentially reducing the number of denials due to easily identifiable factors and improving the overall quality of applications.

4. Monitor and Re-evaluate Model Performance Regularly:

Recommendation: Continuously monitor the performance of the deployed Random Forest model.
Action: Establish a process for periodic re-evaluation of the model's accuracy, precision, recall, and other relevant metrics using new data. Retrain the model as needed to ensure it remains accurate and reflects any changes in visa approval patterns or regulations.
Benefit: Ensures the prioritization system and profile recommendations remain effective and reliable over time.

5. Investigate Outliers in Key Features:

Recommendation: Further investigate applications with extreme values in 'no_of_employees' and 'prevailing_wage' that were identified as outliers during preprocessing.
Action: Analyze these cases manually to understand if there are specific circumstances or data entry issues contributing to these outliers and their impact on case status.
Benefit: Provides deeper insights into edge cases and potentially identifies areas for data quality improvement or specific review protocols.

6. Explore Regional Employment Nuances:

Recommendation: While the model considers 'region_of_employment', a deeper qualitative analysis of regional-specific requirements or trends might be beneficial.
Action: Conduct targeted reviews of applications from regions with slightly different approval rates or unique economic landscapes to identify any uncaptured factors that could further refine recommendations.
Benefit: Allows for more nuanced and region-specific guidance for employers and applicants.

By implementing these recommendations, EasyVisa can leverage the power of data science to create a more efficient, transparent, and effective visa application and approval process, ultimately benefiting both employers seeking talent and foreign workers aiming to contribute to the US economy.